# Harmonized Landsat Sentinel-2 (HLS) Vegetation Indices (VI) Product User Guide

*Product Version 2.0*

Brad Baker, Madhu Sridhar, Junchang Ju, Qiang Zhou, Chris Holden, Chistopher Neigh

Principal Investigator: Christopher Neigh, NASA/GSFC
Correspondence email address: christopher.s.neigh@nasa.gov

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**Acronyms**

| | |
|---|---|
| COG | Cloud Optimized GeoTIFF |
| EVI | Enhanced Vegetation Index |
| MGRS | Military Grid Reference System |
| MSAVI | Modified Soil Adjusted Vegetation Index |
| MSI | Multi-Spectral Instrument |
| NBR | Normalized Burn Ratio |
| NDVI | Normalized Difference Vegetation Index |
| NDWI | Normalized Difference Water Index |
| NIR | Near-Infrared |
| OLI | Operational Land Imager |
| SAVI | Soil Adjusted Vegetation Index |
| SR | Surface Reflectance |
| SWIR | Shortwave Infrared |
| TIRS | Thermal Infrared Sensor |
| TVI | Triangular Vegetation Index |
| UTC | Coordinated Universal Time |
| UTM | Universal Transverse Mercator |
| VI | Vegetation Index |
| WRS-2 | World Reference System-2 |

# 1 Introduction

Vegetation indices are used in Earth Observation to assess and monitor the health and vitality of vegetation across landscapes. These indices combine specific bands of satellite data to quantify various aspects of vegetation, such as the overall biomass. Analysis of vegetation indices allows for tracking changes in vegetation over time, identifying areas of stress or deforestation, and assessing crop health. Vegetation indices provide a reliable and efficient means of understanding the complex dynamics of vegetation health. Under the general term vegetation index, a few other spectral indices not directly related to vegetation health are also included, such as the Normalized Difference Water Index (McFeeters 1996).

## 1.1 HLS overview

The Harmonized Landsat and Sentinel-2 (HLS) project is a collaborative effort by NASA and the USGS to generate consistent surface reflectance (SR) data from a combined array of satellite sensors, namely the Operational Land Imager (OLI) on Landsat-8 and 9 and the Multi-Spectral Instrument (MSI) on Sentinel-2A, 2B, and 2C (which replaced S2A in January 2025). This integration allows for global land monitoring every 2-3 days with a moderate spatial resolution of 30 meters.

The HLS project employs various algorithms, including atmospheric correction, cloud and cloud-shadow masking, common gridding, illumination and view angle normalization, and spectral bandpass adjustment, to create unified products from Landsat and Sentinel-2 data. These HLS data products serve as foundational elements for a "data cube," enabling users to analyze any given pixel over time and to consider the near-daily reflectance time series as if it were collected by a single sensor.

The HLS suite contains both L30 and S30 products, which are derived from Landsat L1TP (Collection 2) and Sentinel-2 L1C input, respectively. These products are gridded into the same Military Grid Reference System (MGRS) tiles with a pixel size of 30 meters. Detailed information can be found in the Harmonized Landsat Sentinel-2 (HLS) Product User Guide version 2.0.

## 1.2 HLS Vegetation (VI) products overview

HLS-VI products are derived from HLS-L30 and S30 surface reflectance products, provided as band separated Cloud Optimized GeoTIFF (COG) files. These products include nine vegetation indices, defined as combinations of various spectral bands.

Each of the HLS-VI L30 and HLS-VI S30 products is generated from the corresponding HLS L30 or HLS S30 product. HLS-VI products have the same pixel resolution, map projection, and spatial extent of L30 and S30. This means that HLS-VI products are produced at a global scale at 30 meter

spatial resolution with a Universal Transverse Mercator (UTM) projection and on the MGRS tiling system.

The HLS VI data processing involves several systematic steps aimed at ensuring accuracy and contextual understanding. First, if a pixel has a non-positive surface reflectance in any of the bands listed in Table 1, it is assigned a fill value of -19,999 for the vegetation index (VI). The HLS Quality Assessment (QA) band, specifically the Fmask (Qiu et al. 2019), is used to mask out pixels classified as cloud, cloud shadow, or adjacent to cloud or cloud shadow, and the VI fill value is assigned accordingly. Next, vegetation indices are computed using specific algorithms tailored to the spectral bands, as detailed in Table 1. A VI metadata file is generated using metadata attributes from the HLS granule, along with attributes specific to the VI granule. Notably, the metadata file includes the URL to the original HLS L30/S30 QA band. All VI granules are saved in the int16 data type to maintain consistency and reduce data storage. While most VI granules are scaled by a factor of 10000 to save as int16, the Triangular Vegetation Index (TVI) is scaled by 100.

### 1.3 Spectral reflectance bands used to generate HLS-VI products

The spectral reflective bands from Landsat-8 and 9 OLI and Sentinel-2 and 2A MSI used to generate HLS-VI products are described in Table 1.

*Table 1: HLS spectral bands nomenclature*

| **Band Name** | **OLI band number** | **MSI band number** | **HLS band code name L8** | **HLS band code name S2** | **Wavelength (micrometers )** |
|---|---|---|---|---|---|
| Blue | 2 | 2 | B02 | B02 | 0.45-0.51* |
| Green | 3 | 3 | B03 | B03 | 0.53-0.59* |
| Red | 4 | 4 | B04 | B04 | 0.64-0.67* |
| NIR | 5 | 8A | B05 | B8A | 0.78-0.88** |
| SWIR 1 | 6 | 11 | B06 | B11 | 1.57-1.65* |
| SWIR 2 | 7 | 12 | B07 | B12 | 2.11-2.29* |

*from OLI specifications
*from MSI specifications

## 2 HLS-VI products

This section describes the nine vegetation indices (Table 2) generated using HLS L30 and S30 data products.

*Table 2: VI and the associated equations used to create the HLS VI product.*

| **Index** | **L30** | **S30** |
|---|---|---|
| Normalized Difference Vegetation Index (NDVI) | $\frac{B5 - B4}{B5 + B4}$ | $\frac{B8A - B4}{B8A + B4}$ |
| Enhanced Vegetation Index (EVI) | $\frac{2.5 * (B5 - B4)}{B5 + 6 * B4 - 7.5 * B2 + 1}$ | $\frac{2.5 * (B8A - B4)}{B8A + 6 * B4 - 7.5 * B2 + 1}$ |
| Soil Adjusted Vegetation Index (SAVI) | $1.5 * \frac{B5 - B4}{B5 + B4 + 0.5}$ | $1.428 * \frac{B8A - B4}{B8A + B4 + 0.428}$ |
| Modified Soil Adjusted Vegetation Index (MSAVI) | $\frac{2 * B5 + 1 - \sqrt{(2 * B5 + 1)^2}}{2}$ | $\frac{2 * B8A + 1 - \sqrt{(2 * B8A + 1)^2}}{2}$ |
| Normalized Difference Moisture Index (NDMI) | $\frac{B5 - B6}{B5 + B6}$ | $\frac{B8A - B11}{B8A + B11}$ |
| Normalized Difference Water Index (NDWI) | $\frac{B3 - B5}{B3 + B5}$ | $\frac{B3 - B8A}{B3 + B8A}$ |
| Normalized Burn Ratio (NBR) | $\frac{B5 - B7}{B5 + B7}$ | $\frac{B8A - B12}{B8A + B12}$ |
| Normalized Burn Ratio 2 (NBR2) | $\frac{B6 - B7}{B6 + B7}$ | $\frac{B11 - B12}{B11 + B12}$ |
| Triangular Vegetation Index (TVI) | $\frac{120(B5 - B3) - 200(B4 - B}{2}$ | $\frac{120(B8A - B3) - 200(B4 - B3}{2}$ |

## 2.1 Normalized Difference Vegetation Index

The Normalized Difference Vegetation Index (NDVI) serves as a ubiquitous tool in remote sensing, offering insights into vegetation presence and vitality. By combining near-infrared (NIR)

and red reflectance, NDVI provides indications of vegetation density and stress levels, proving invaluable for environmental monitoring and vegetation assessment (Tucker 1979).

## 2.2 Enhanced Vegetation Index

The Enhanced Vegetation Index (EVI) was developed to mitigate atmospheric influences and increase sensitivity to high biomass (Huete et al. 2002). EVI is particularly useful in regions with dense vegetation cover or significant atmospheric interference.

## 2.3 Soil Adjusted Vegetation Index

In regions with high soil reflectance, the Soil Adjusted Vegetation Index (SAVI) diminishes soil brightness effects on vegetation detection (Huete 1988). By introducing a soil adjustment factor, SAVI enhances vegetation detection sensitivity, especially in arid and semi-arid areas where soil reflectance can compromise index accuracy.

## 2.4 Modified Soil Adjusted Vegetation Index

An adaptation of SAVI, the Modified Soil Adjusted Vegetation Index (MSAVI) further improves vegetation detection accuracy, particularly in areas with dense vegetation or high soil reflectance (Qi et al. 1994). By addressing soil brightness and saturation effects from abundant vegetation cover, MSAVI emerges as a reliable indicator of vegetation health across various ecosystems.

## 2.5 Normalized Difference Moisture Index

Developed to quantify vegetation moisture content or water stress, the Normalized Difference Moisture Index (NDMI) leverages the difference between NIR and shortwave infrared (SWIR) reflectance (Gao 1996). NDMI's ability to discern moisture content makes it instrumental in monitoring vegetation health and water availability.

## 2.6 Normalized Difference Water Index

The Normalized Difference Water Index (NDWI) is a remote sensing index used to detect the presence and changes in water bodies within an image (McFeeters 1996). It measures the difference in NIR and green band reflectance. High NDWI values indicate the presence of water, while low values indicate non-water features like soil or vegetation. NDWI is particularly useful for monitoring changes in surface water bodies, such as lakes, rivers, and wetlands, and for assessing water-related environmental dynamics.

## 2.7 Normalized Burn Ratio

The Normalized Band Ratio (NBR) is used to assess vegetation health and monitor disturbances such as wildfires (Key & Benson 2005). It is calculated as the normalized difference between the NIR and SWIR1 bands. Positive NBR values typically indicate healthy vegetation, while negative

values suggest burned or stressed vegetation. NBR is valuable for assessing changes in vegetation conditions, particularly in areas affected by fire, drought, or other environmental disturbances.

### 2.8 Normalized Burn Ratio 2

Developed for assessing wildfire burn severity, the Normalized Burn Ratio 2 (NBR2) measures the disparity between SWIR1 and SWIR2 reflectance. By offering assessments of vegetation health and burn severity, NBR2 provides gradations of damage severity, aiding in wildfire impact assessments.

### 2.9 Triangular Vegetation Index

The Triangular Vegetation Index (TVI) is a vegetation index derived from remote sensing data, particularly from multispectral imagery (Broge & Leblanc 2001). It is calculated using the red, green, and blue bands to estimate vegetation health and vigor. TVI enhances the contrast between vegetation and soil, making it useful for monitoring changes in plant biomass and stress levels. It's particularly effective in assessing vegetation health in agricultural fields and natural ecosystems.

## 3 Spatial Coverage

HLS-VI v2.0 encompasses all global land except Antarctica, as shown in a land mask (Fig. 1) derived from the NOAA shoreline dataset (http://www.ngdc.noaa.gov/mgg/shorelines/data/gshhg/latest/). Antarctica is excluded due to low solar elevations which compromise the plane-parallel atmospheric correction. Additionally, note that data acquisition over some small oceanic islands by the Landsat and Sentinel-2 sensors may not be made regularly.

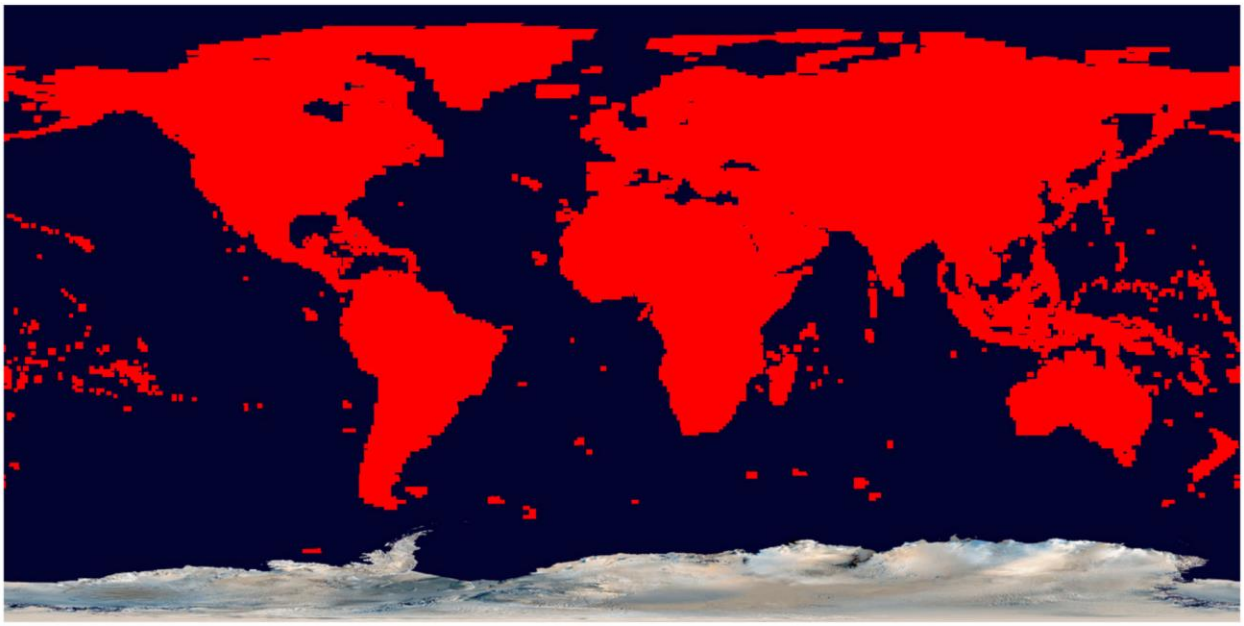

*Figure 1: HLS VI v2.0 covers the global land, including major islands but excluding Antarctica.*

# 4 Product Formats

## 4.1 File format

HLS-VI 2.0 products are in Cloud Optimized GeoTIFF (COG), one file per data layer to offer the flexibility of only downloading the needed data layers and, for cloud-based applications, the needed spatial subsets within a tile. The COG files are internally compressed. L30 VI data are stored in directories such as HLS-VI.L30.T17SLU.2020209T155956.v2.0, which suggests the source reflectance from L30 over tile 17SLU from data acquired on day 209 of 2020 specifically at UTC 155956. This example product consists of the following files:

HLS-VI.L30.T17SLU.2020209T155956.v2.0.NDVI.tif
HLS-VI.L30.T17SLU.2020209T155956.v2.0.EVI.tif
HLS-VI.L30.T17SLU.2020209T155956.v2.0.SAVI.tif
HLS-VI.L30.T17SLU.2020209T155956.v2.0.MSAVI.tif
HLS-VI.L30.T17SLU.2020209T155956.v2.0.NDMI.tif
HLS-VI.L30.T17SLU.2020209T155956.v2.0.NBR.tif
HLS-VI.L30.T17SLU.2020209T155956.v2.0.NBR2.tif
HLS-VI.L30.T17SLU.2020209T155956.v2.0.TVI.tif
HLS-VI.L30.T17SLU.2020209T155956.v2.0.cmr.xml
HLS-VI.L30.T17SLU.2020209T155956.v2.0.NDVI.jpeg

The filenames for individual VI bands are self-explanatory. File HLS-VI.L30.T17SLU.2020209T155956.v2.0.cmr.xml is the metadata file and HLS-

VI.L30.T17SLU.2020209T155956.v2.0.NDVI.jpeg is an NDVI browse image. We chose NDVI as a representative VI granule for the browse image. The UTC time in the filenames is the sensing time at the input Landsat 8 and 9 scene centers. After gridding into the MGRS tiles it does not accurately indicate the sensing time over the tile. If two scenes overlap a MGRS tile, the sensing time of one of the scenes is chosen by chance. So, this timing information is not accurate for the MGRS tile center; it is intended mainly as an identifier, not for quantitative analysis. The metadata file (cmr.xml) contains the URL to the original L30/S30 product's QA file, which can be used if additional filtering of water or snow or high aerosol pixels is desired.

S30-VI data are stored in the same format. An example directory HLS-VI.S30.T17SLU.2020117T160901.v2.0 may contain the following files:

HLS-VI.S30.T17SLU.2020117T160901.v2.0.NDVI.tif
HLS-VI.S30.T17SLU.2020117T160901.v2.0.EVI.tif
HLS-VI.S30.T17SLU.2020117T160901.v2.0.SAVI.tif
HLS-VI.S30.T17SLU.2020117T160901.v2.0.MSAVI.tif
HLS-VI.S30.T17SLU.2020117T160901.v2.0.NDMI.tif
HLS-VI.S30.T17SLU.2020117T160901.v2.0.NDWI.tif
HLS-VI.S30.T17SLU.2020117T160901.v2.0.NBR.tif
HLS-VI.S30.T17SLU.2020117T160901.v2.0.NBR2.tif
HLS-VI.S30.T17SLU.2020117T160901.v2.0.TVI.tif
HLS-VI.S30.T17SLU.2020117T160901.v2.0.cmr.xml
HLS-VI.S30.T17SLU.2020117T160901.v2.0.NDVI.jpeg

The UTC time in the S30 product filenames is the time the sensor begins to sense the sunlit side of the Earth for each orbit, not the exact sensing time over the tile center. When a sequence of observations is available on the same day at the high latitude, they can still be differentiated by the timing information.

## 4.2 L30 and S30 VI layers

The HLS-VI L30 and S30 products contain nine vegetation indices gridded at 30 m resolution in MGRS tiles. Table 3 describes the properties of the VI layers which are the same for both L30 and S30 products. Note that the scale factor for the TVI layer is 0.01 whereas all the other VI layers have a scale factor of 0.0001.

*Table 3: L30-VI and S30-VI layers*

| Data layer | Units | Data type | Scale | Fill value | Valid Range | Spatial resolution (meters) |
|---|---|---|---|---|---|---|
| NDVI | none | int16 | 0.0001 | -19,999 | -1 to 1 | 30 |
| EVI | none | int16 | 0.0001 | -19,999 | -1 to 1 | 30 |
| SAVI | none | int16 | 0.0001 | -19,999 | -1 to 1 | 30 |
| MSAVI | none | int16 | 0.0001 | -19,999 | -1 to 1 | 30 |
| NDMI | none | int16 | 0.0001 | -19,999 | -1 to 1 | 30 |
| NDWI | none | int16 | 0.0001 | -19,999 | -1 to 1 | 30 |
| NBR | none | int16 | 0.0001 | -19,999 | -1 to 1 | 30 |
| NBR2 | none | int16 | 0.0001 | -19,999 | -1 to 1 | 30 |
| TVI | none | int16 | 0.01 | -19,999 | NA | 30 |

## 4.4 Metadata

Metadata about the L30-VI and S30-VI products are presented in the cmr.xml file and in the GeoTIFF files.

4.3.1 Key metadata for L30-VI and S30-VI granules from the complete XML list include:

- GRANULE_UR

The VI granule product ID

- ACCODE

The version of LaSRC used by HLS for L30 or S30

- HORIZONTAL_CS_NAME

The map projection of the input Landsat scene or scenes. The UTM zone of the input Landsat scene may be different from that of L30.

- MEAN_SUN_ZENITH_ANGLE

The mean solar zenith in the tile

- MEAN_SUN_AZIMUTH_ANGLE

The mean solar azimuth in the tile

- MEAN_VIEW_ZENITH_ANGLE

The mean view zenith in the tile

- MEAN_VIEW_AZIMUTH_ANGLE

The mean view azimuth in the tile

- NBAR_SOLAR_ZENITH

The solar zenith angle used in NBAR derivation.

- MGRS_TILE_ID

The unique identifier assigned to a specific geographic area.

- SENSING_TIME

The WRS scene center sensing time, carried over from the Level-1 metadata; not precisely representing the data gridded into the tile. When two scenes overlap the tile, the sensing time for each is retained.

- INSTRUMENT

Instrument used to measure radiance

- PLATFORM

The platform that carries the sensor

- SPATIAL_COVERAGE

The percentage of the tile with data

- CLOUD_COVERAGE

The percentage of cloud and cloud shadow in observation based on Fmask

4.3.2 Metadata included in L30-VI and S30-VI GeoTIFF files:

- ACCODE

The version of LaSRC used by HLS for L30

- ADD_OFFSET

Value added to the VI data before they are scaled to int16 reflectance data

- AREA_OR_POINT

The type of sampling for each pixel

- CLOUD_COVERAGE

The percentage of cloud and cloud shadow in observation based on Fmask

- HLS-VI_PROCESSING_TIME

The time that the VI granule was generated

- HORIZONTAL_CS_NAME

The map projection of the input Landsat scene or scenes. The UTM zone of the input Landsat scene may be different from that of L30.

- LONGNAME

The long name of the vegetation index

- MEAN_SUN_AZIMUTH_ANGLE

The mean solar azimuth in the tile

- MEAN_SUN_ZENITH_ANGLE

The mean solar zenith in the tile

- MEAN_VIEW_AZIMUTH_ANGLE

The mean view azimuth in the tile

- MEAN_VIEW_ZENITH_ANGLE

The mean view azimuth in the tile

- NBAR_SOLAR_ZENITH

The solar zenith angle used in NBAR derivation.

- NCOLS and NROWS

The number of columns and rows in the tile

- SATELLITE

The platform that carries the sensor

- SCALE_FACTOR

Multiplier to be applied to the int16 VI data to get the unscaled value

- SENSING_TIME

The WRS scene center sensing time, carried over from the Level-1 metadata; not precisely representing the data gridded into the tile. When two scenes overlap the tile, the sensing time for each is retained.

- SPATIAL_COVERAGE

The percentage of the tile with data

- SPATIAL_RESOLUTION

The size of a pixel on the ground

- ULX and ULY

The UTM X/Y coordinate at the upper left corner of the tile

- TILE_ID

The unique MGRS tile identifier assigned to a specific geographic area.

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